

Ongoing Projects

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Ongoing Work

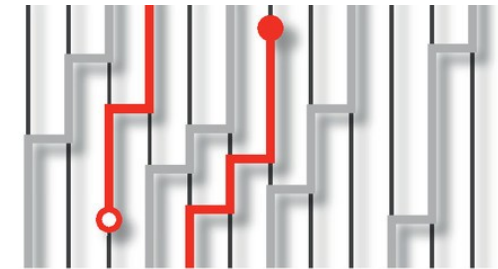
- Methods (DQMC, AC)
- 2D Holstein Model Crossover
- 2D Optical SSH Model Crossover
- Parallel Tempering

SmoQyDQMC.jl



Ben Cohen-Stead

- DQMC – You are at least passingly familiar
 - If not, see Gubernatis, Kawashima, Werner
 - (Probably free in pdf via UC Davis)
- SmoQyDQMC.jl
 - All lattice geometries
 - HMC updates
 - Hubbard + Phonon models



JAMES GUBERNATIS
NAOKI KAWASHIMA
PHILIPP WERNER

QUANTUM
MONTE
CARLO
METHODS

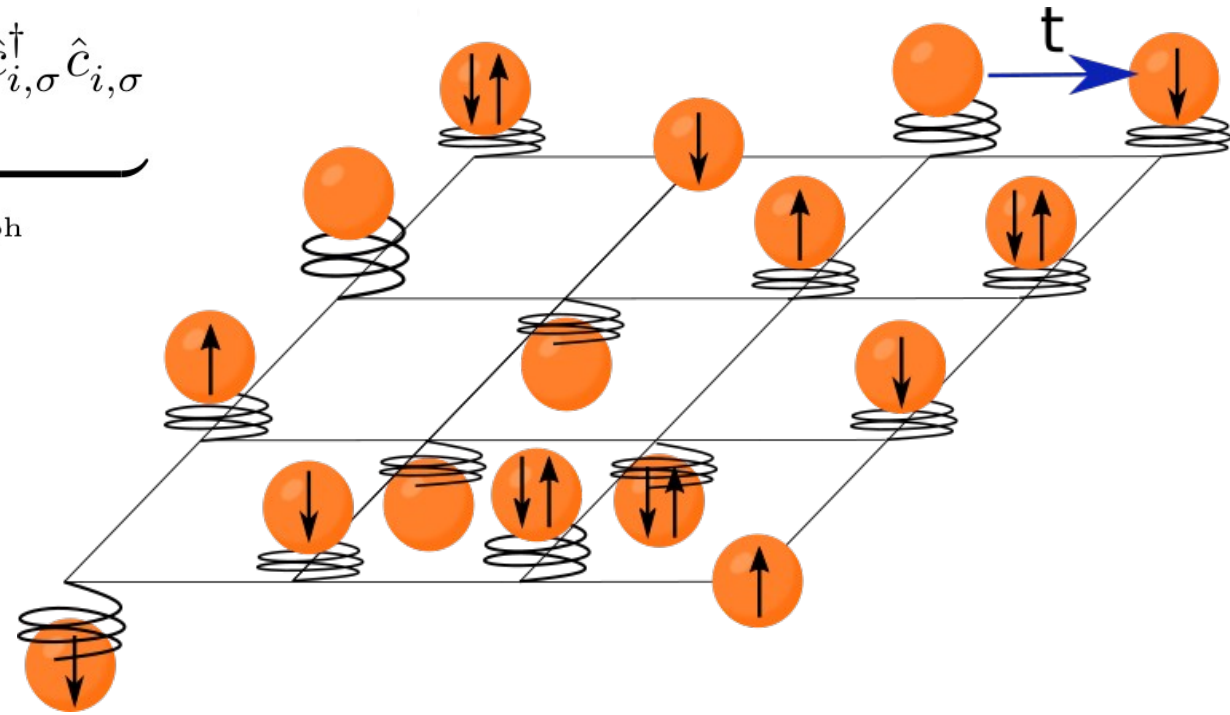
Algorithms for Lattice Models



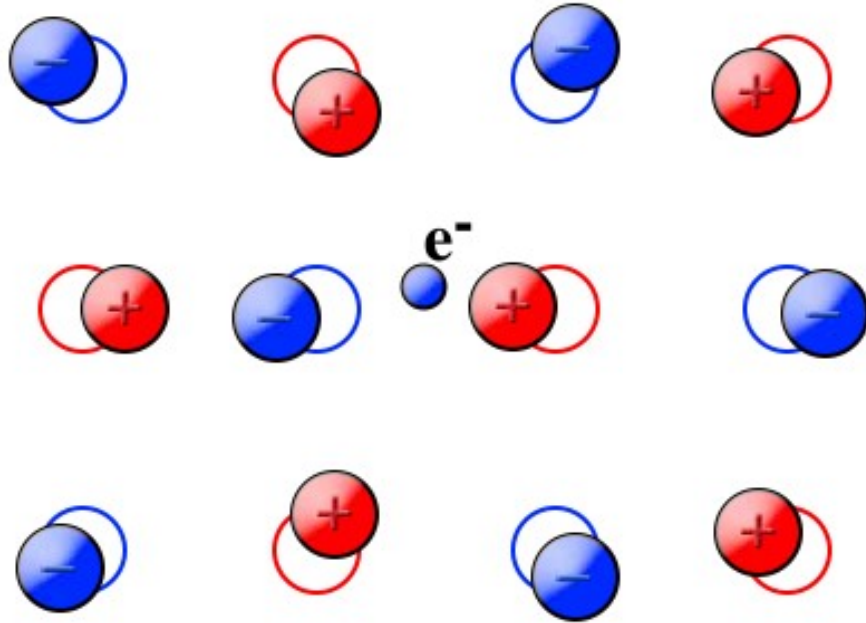
2D Holstein Model

$$H_{\text{Holstein}} = \underbrace{-t \sum_{\sigma, \langle i, j \rangle} \left(\hat{c}_{i, \sigma}^\dagger \hat{c}_{j, \sigma} + \text{h.c.} \right) - \mu \sum_{i, \sigma} \hat{c}_{i, \sigma}^\dagger \hat{c}_{i, \sigma}}_{H_{\text{el}}} + \underbrace{\sum_i \left(\frac{1}{2M} \hat{P}_i^2 + \frac{1}{2} M \Omega^2 \hat{X}_i^2 \right)}_{H_{\text{ph}}} + \underbrace{\alpha \sum_{\sigma, i} \hat{X}_i \hat{c}_{i, \sigma}^\dagger \hat{c}_{i, \sigma}}_{H_{\text{el-ph}}}$$

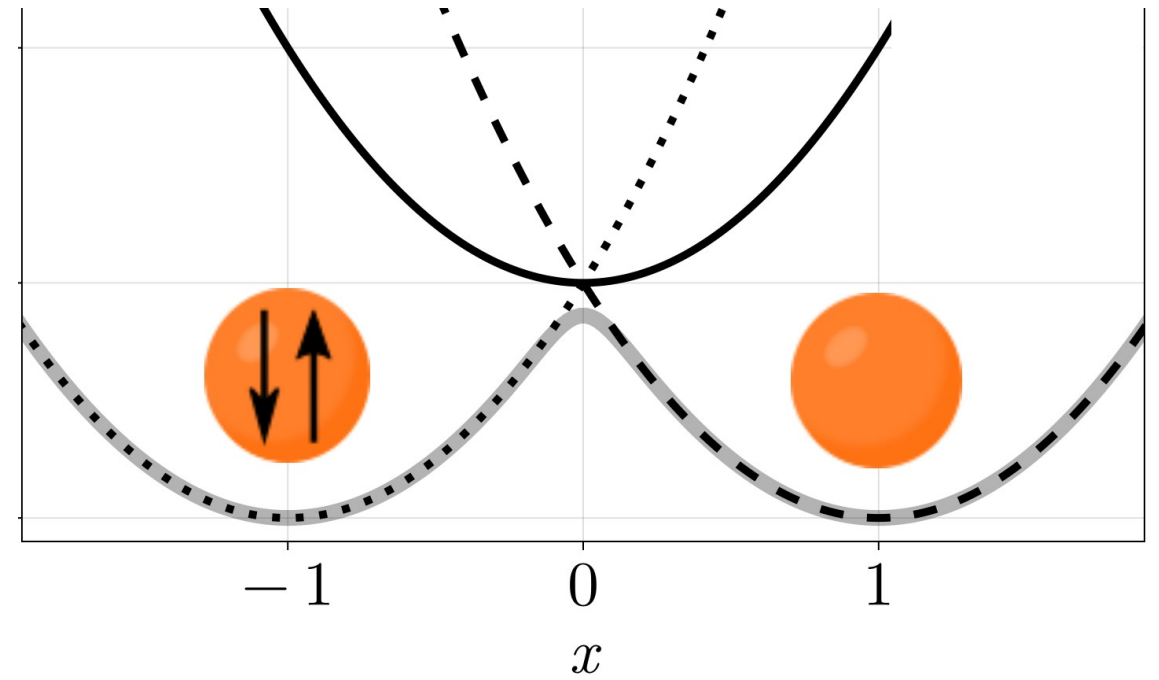
$$\lambda = \alpha^2 / (M \Omega^2 W)$$



Polaron / Bipolaron



$$\alpha \sum_{i\sigma} \hat{X}_i \left(\hat{n}_{i,\sigma} - \frac{1}{2} \right)$$



Measurements

$$S_z(\vec{r}) = \frac{1}{N} \sum_j \langle \hat{S}_z(j + \vec{r}) \hat{S}_z(j) \rangle$$

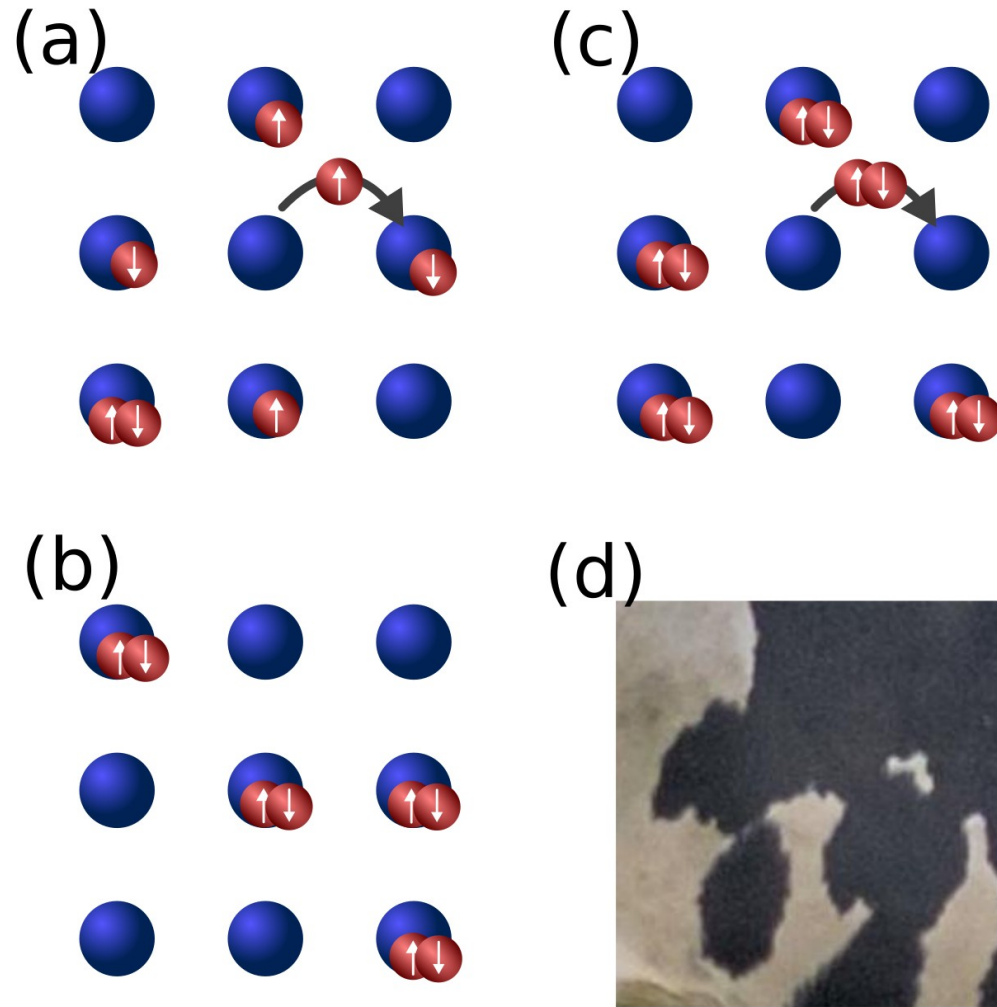
$$S_z(\vec{q}) = \sum_{\vec{r}} e^{-i\vec{q} \cdot \vec{r}} S_z(\vec{r})$$

$$S_{cdw}(\pi, \pi) = \sum_{\vec{j}} (-1)^{j_x + j_y} \langle \hat{n}(j) \hat{n}(0) \rangle$$

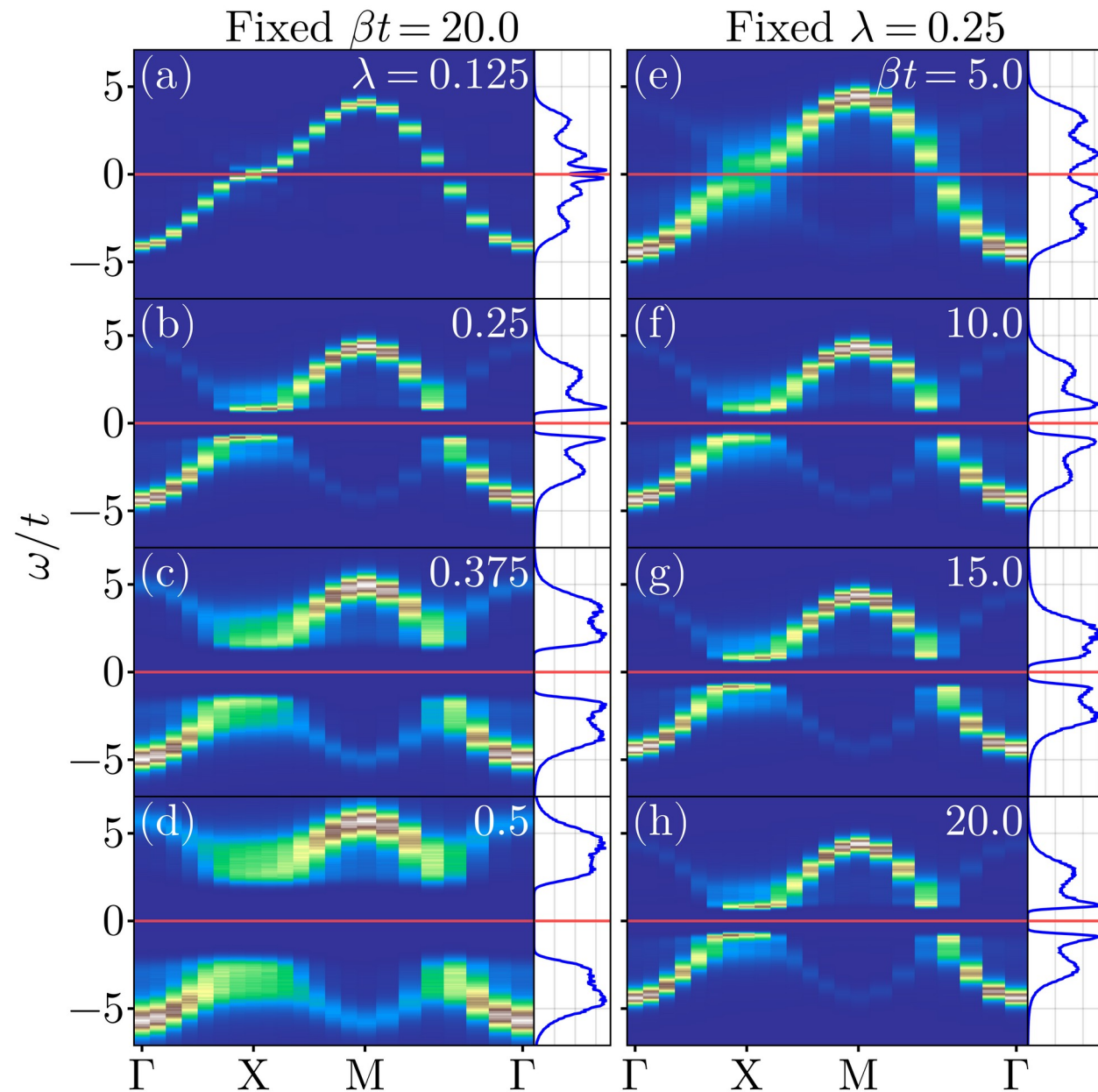
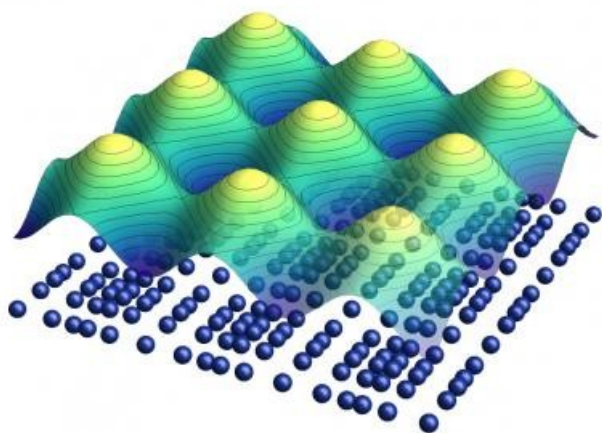
$$G(\vec{r} = 0, \tau) = \int_{-\infty}^{\infty} d\omega \frac{e^{-\tau\omega}}{1 + e^{-\beta\omega}} A_{DOS}(\omega)$$

$$\Pi(\vec{q} = 0, \tau) = \int_{-\infty}^{\infty} d\omega \frac{\omega e^{-\tau\omega}}{1 - e^{-\beta\omega}} \sigma(\omega)$$

Holstein Conduction States

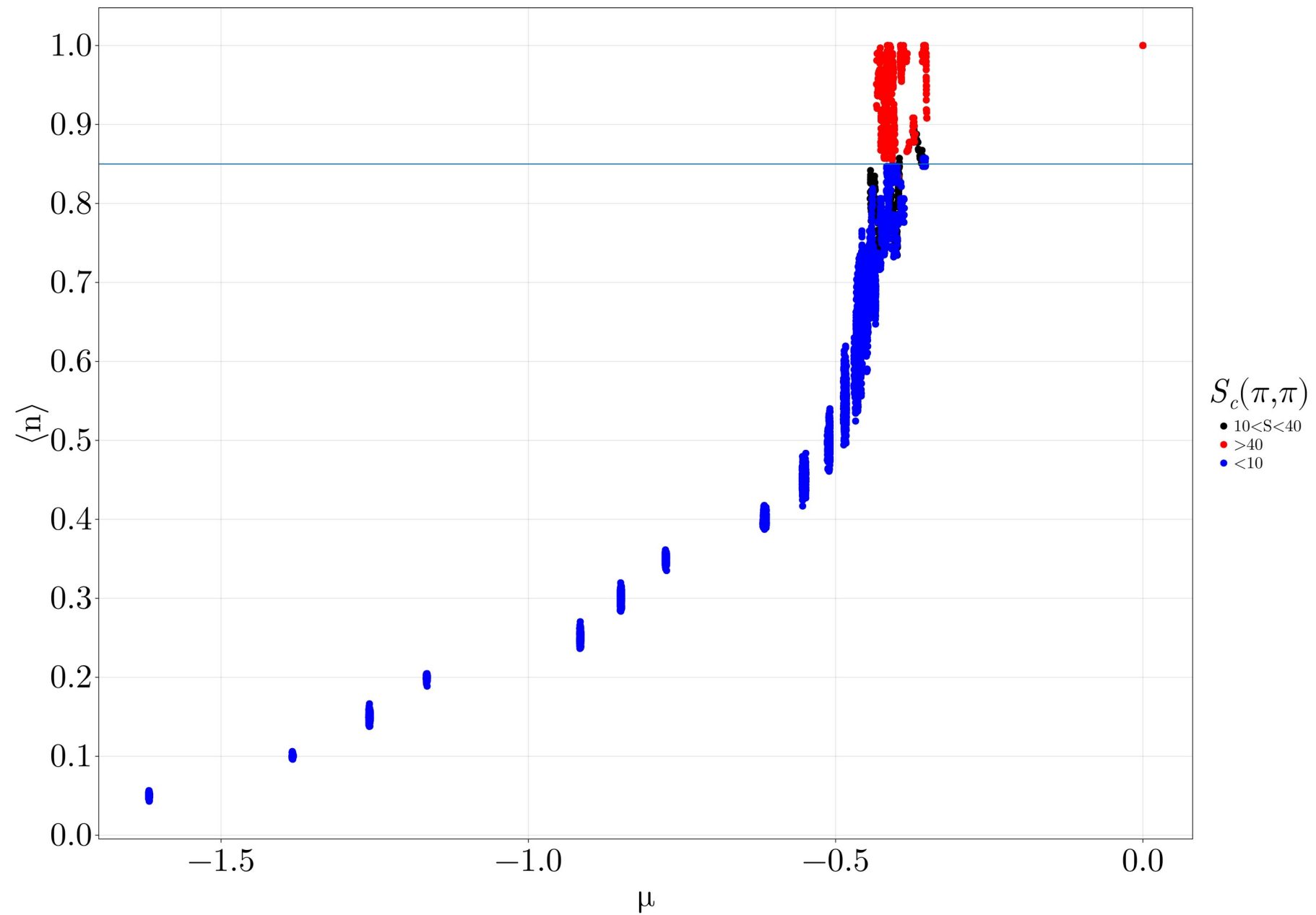


Half Filling Charge Density Wave, $q=(\pi,\pi)$



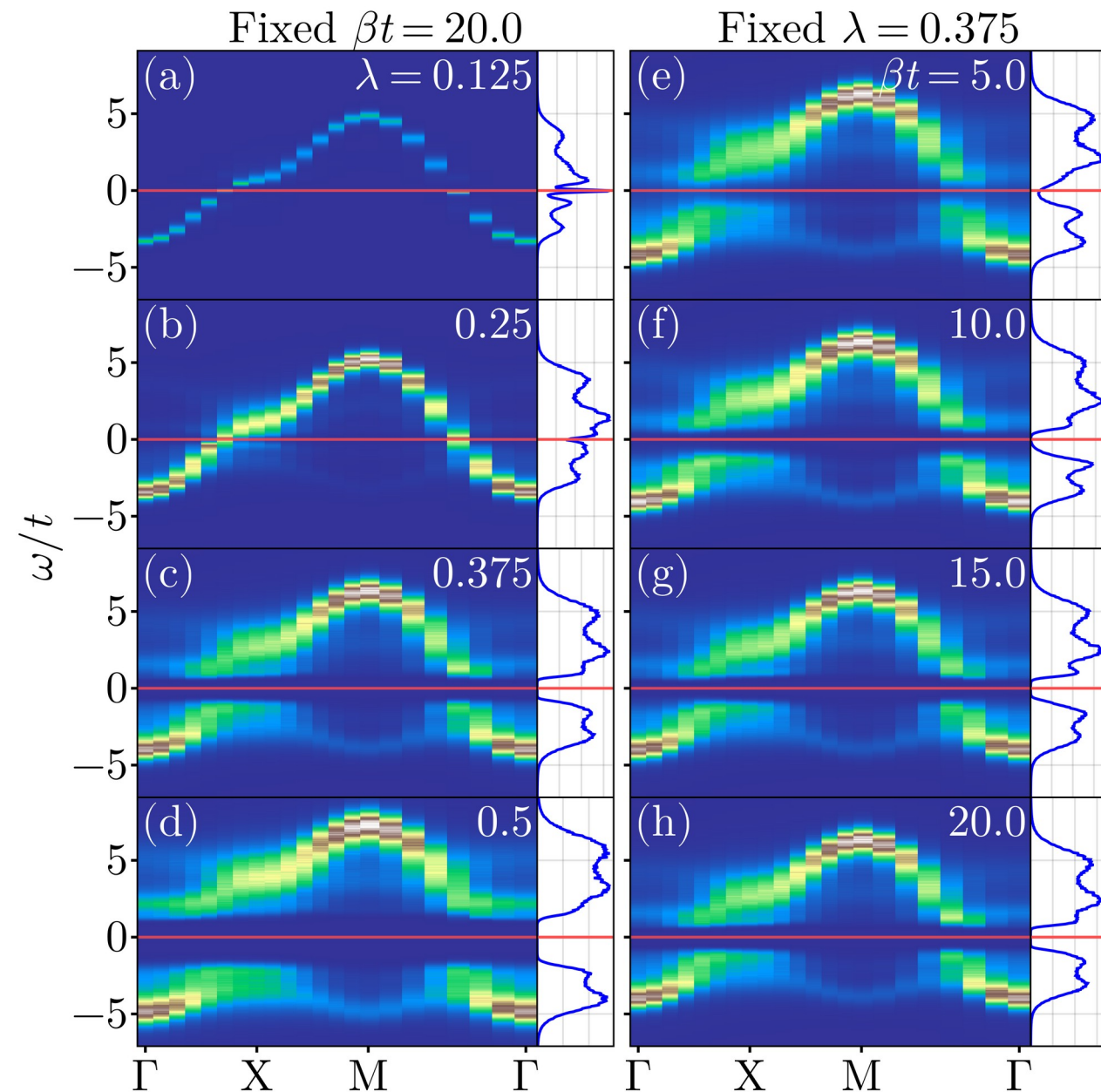
- Filling vs Chemical Potential

- $\lambda=0.3$, $\Omega=0.5t$,
 $\beta t = 20.0$



$$\langle n \rangle = 0.7, \Omega = 0.5t$$

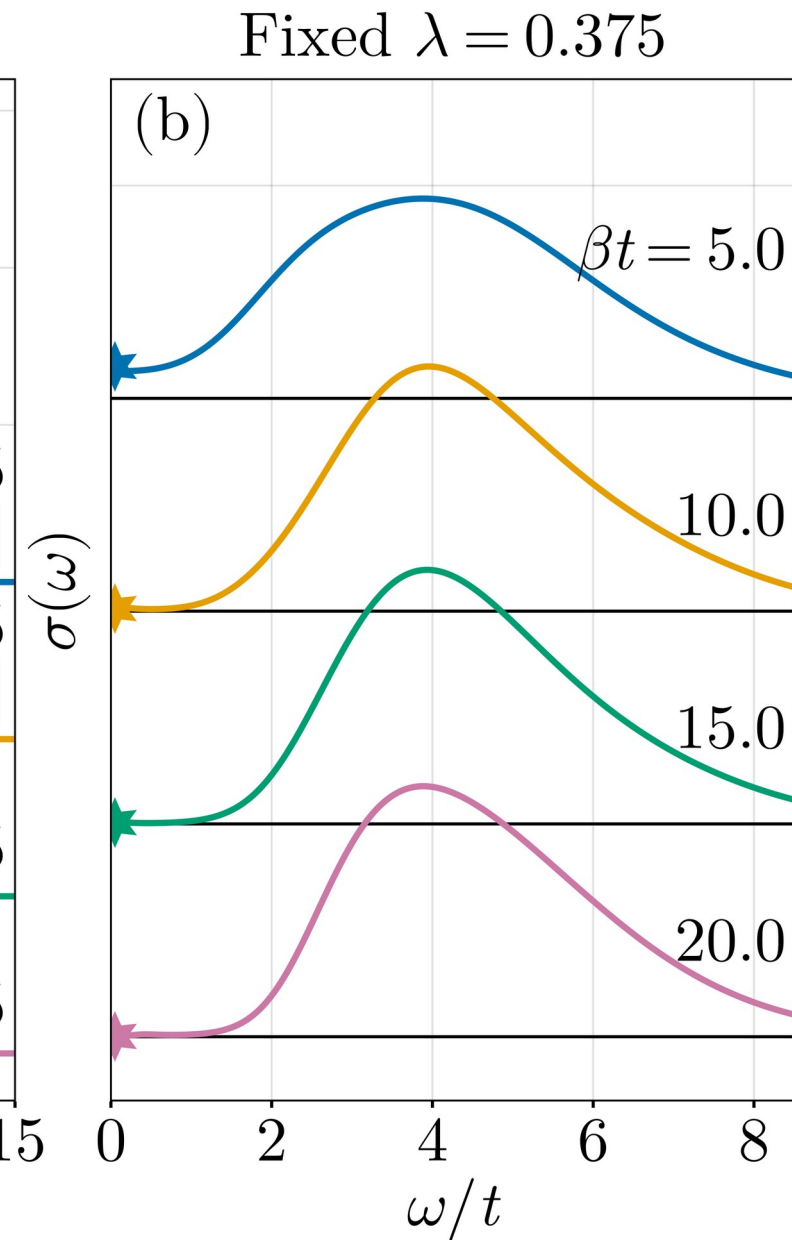
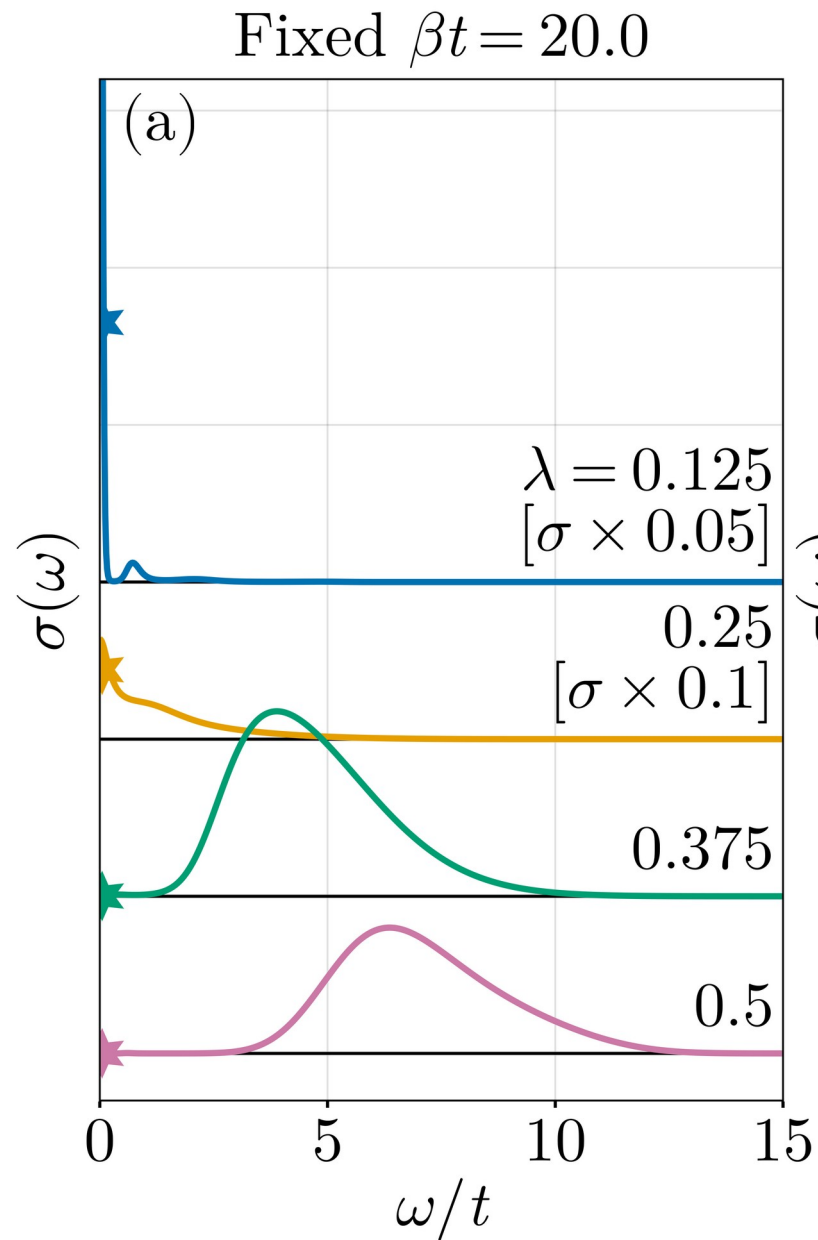
- Mild phase separation
 - Fluctuations in $S_c(\pi, \pi)$
 - Faint backfolding
- Kinked dispersion
- Gapped DOS at low T, medium-high coupling



$$\langle n \rangle = 0.7$$

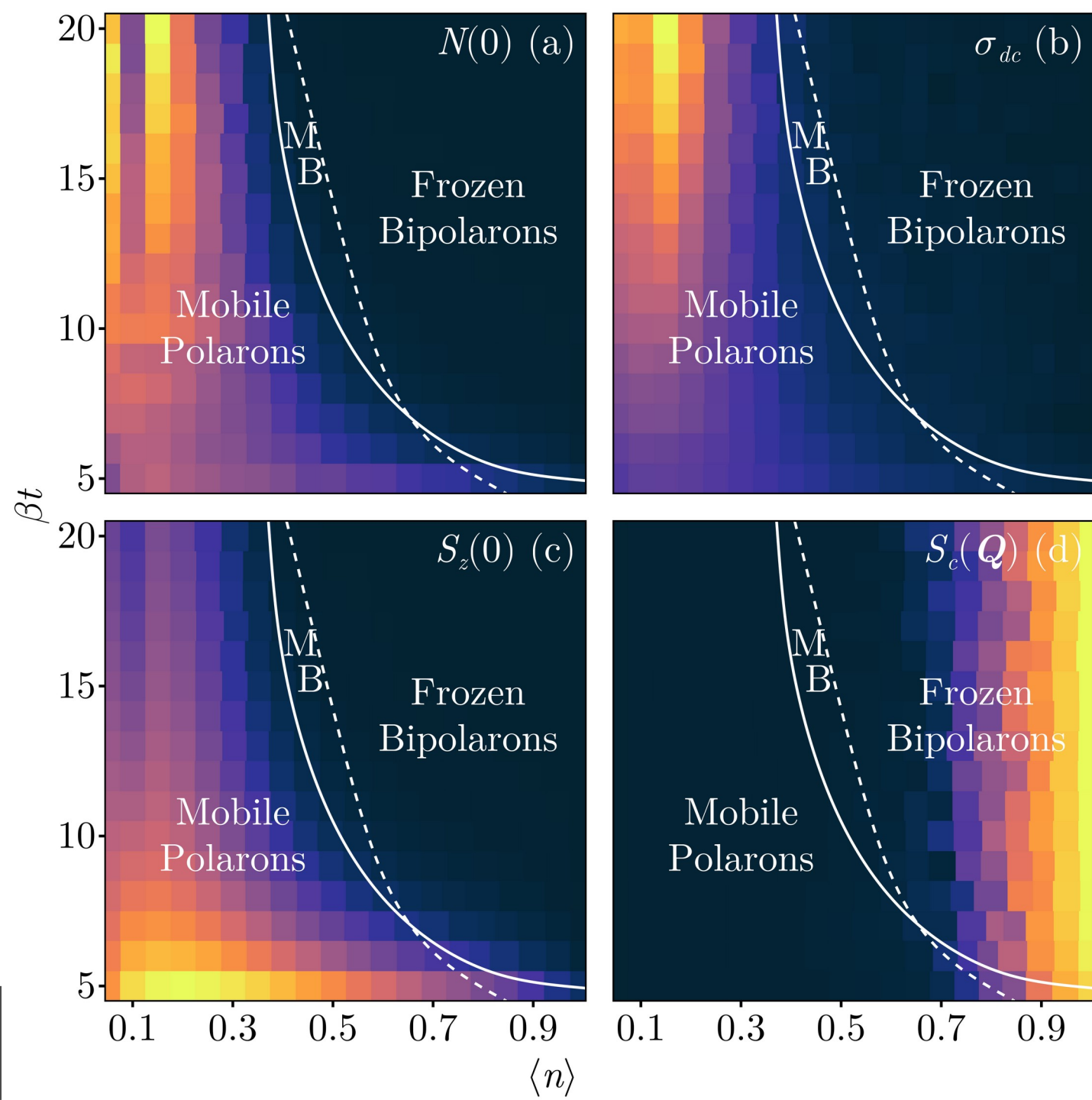
$$\Omega = 0.5t$$

- Tiny (but non zero) DC conductivity for $\beta \leq 15$
-



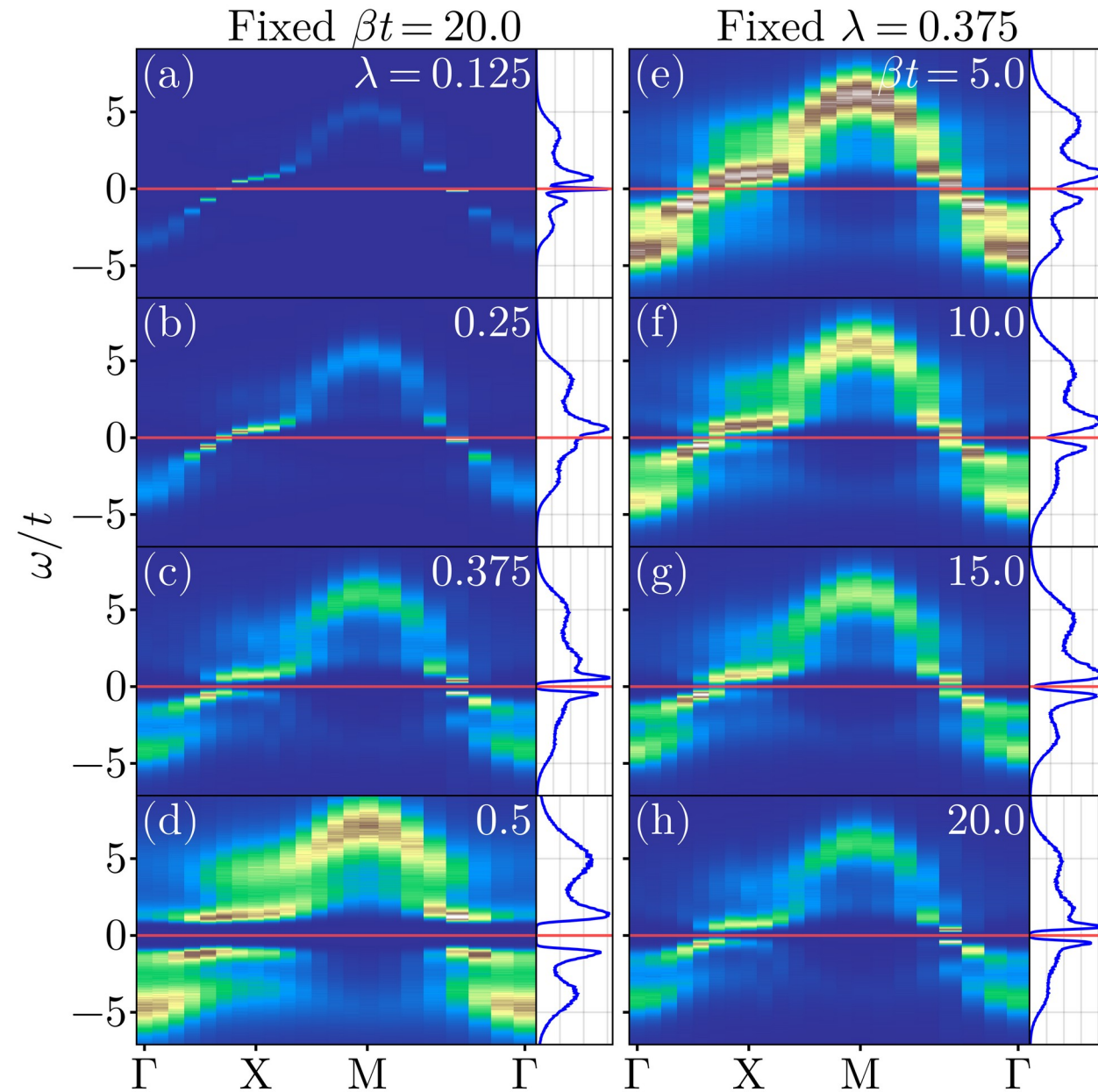
$$\langle n \rangle = 0.7, \Omega = 0.5t$$

- Crossover diagram dominated by mobile polarons and non-conductive bipolarons
- Possible small region of conductive bipolarons
- DC conductivity on Sqrt scale



$$\langle n \rangle = 0.7, \Omega = 2.0t$$

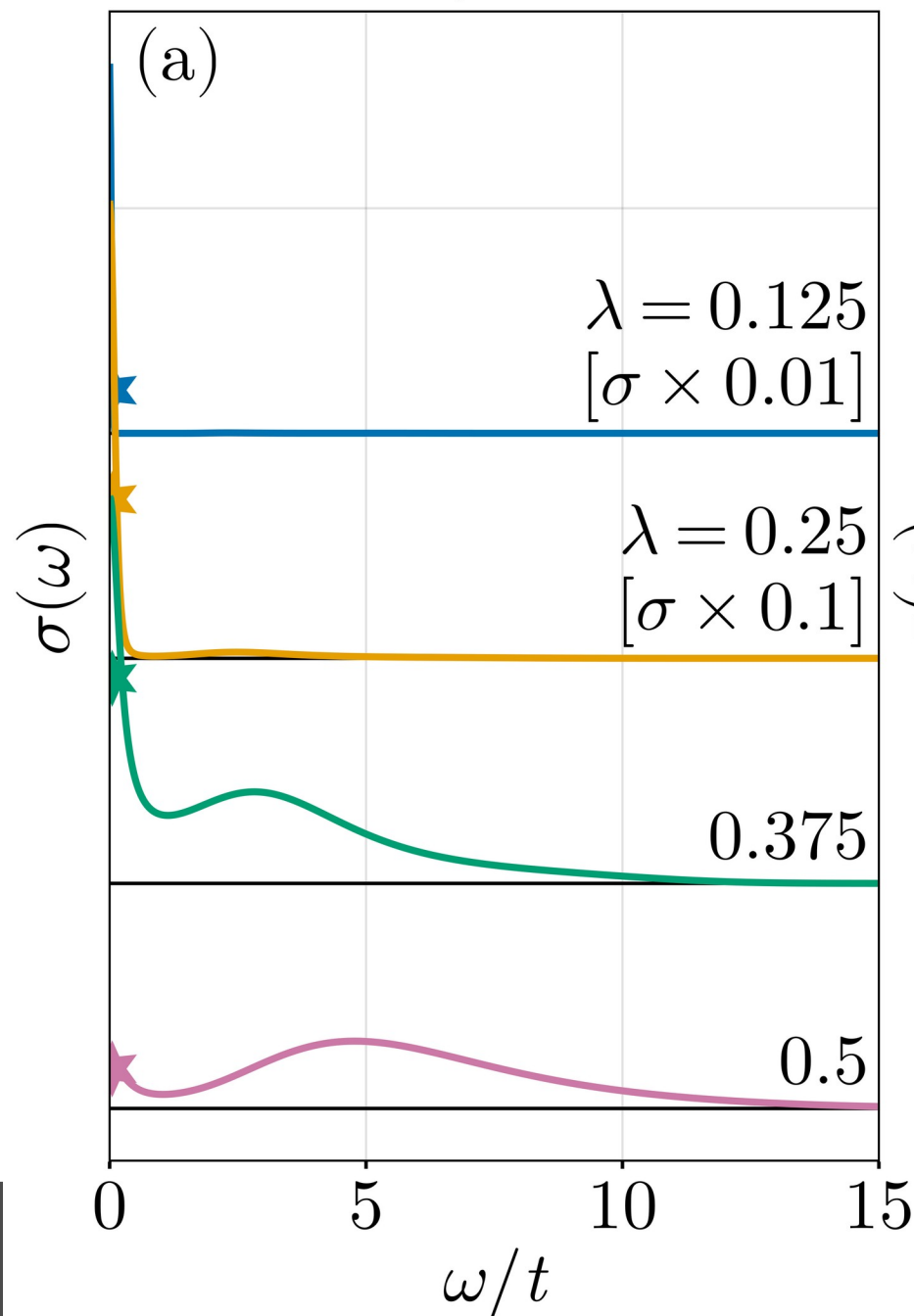
- Mild phase separation
 - Fluctuations in $S_c(\pi, \pi)$
 - Faint backfolding
- Kinked dispersion and clear Franck-Condon effects
- Gapped DOS at low T, medium-high coupling



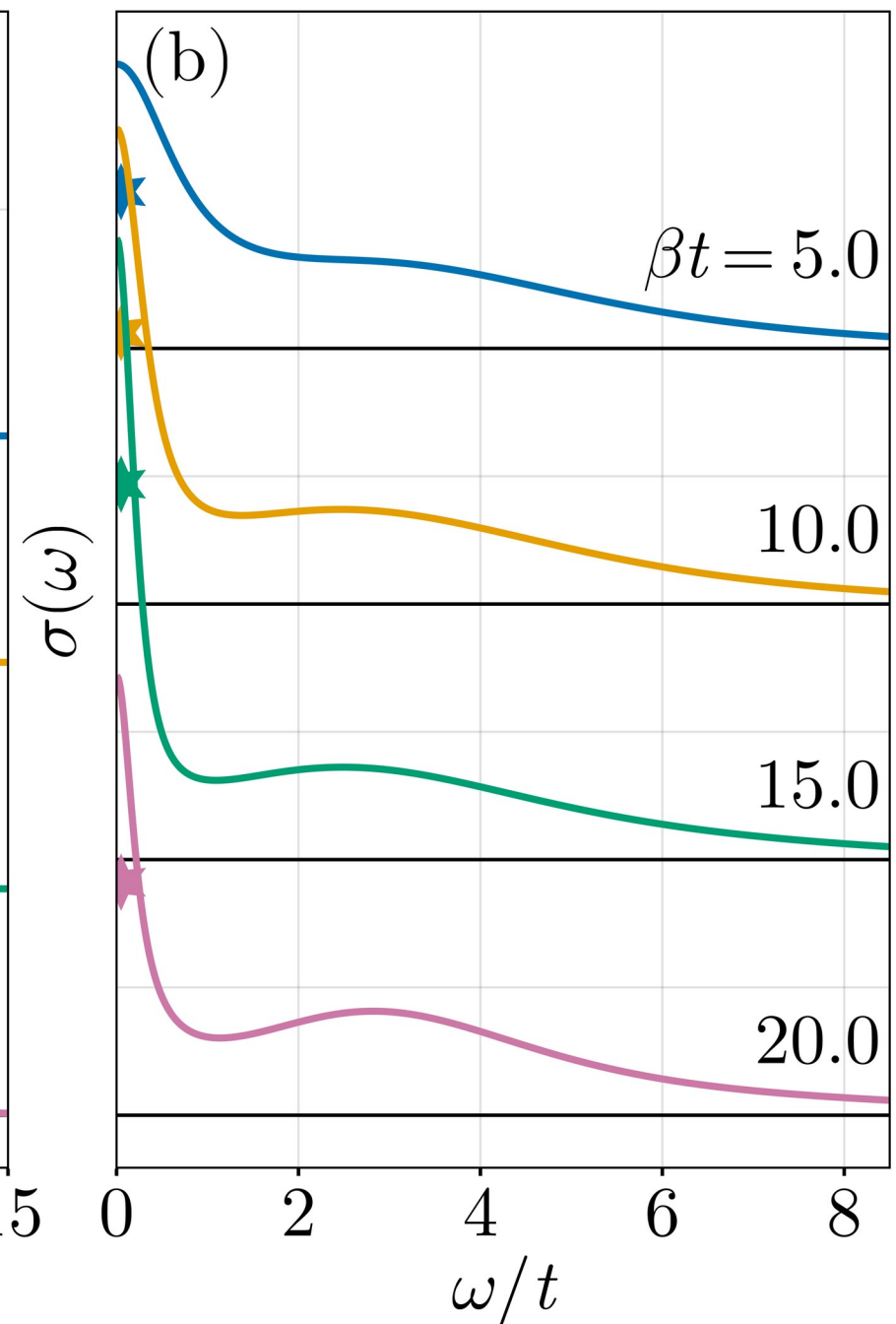
$\langle n \rangle = 0.7$
 $\Omega = 2.0t$

- Clear conductivity
For all plots

Fixed $\beta t = 20.0$

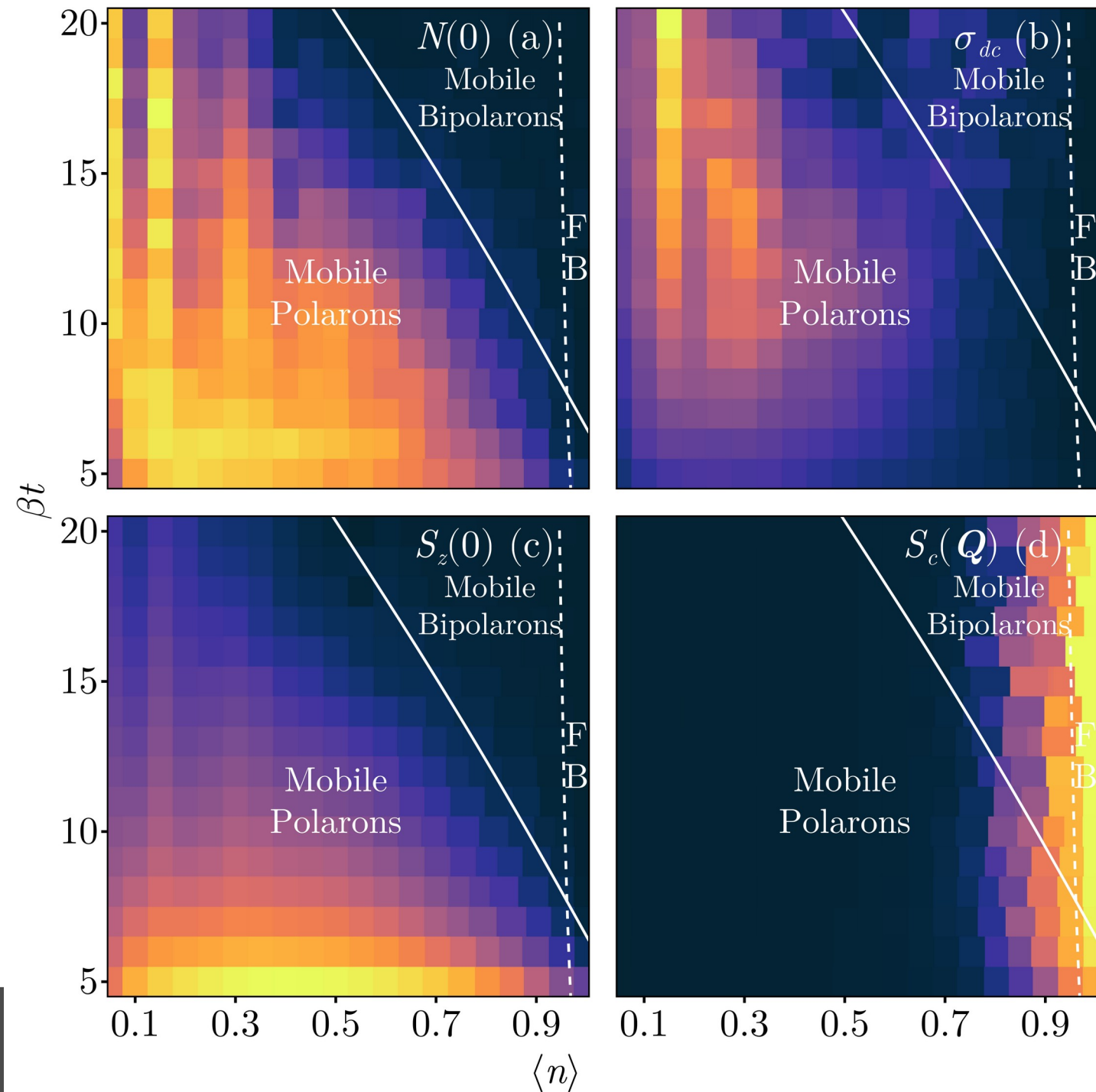


Fixed $\lambda = 0.375$



$$\langle n \rangle = 0.7, \Omega = 2.0t$$

- Clearly mobile bipolarons
- DC conductivity NOT on Sqrt scale



2D Optical SSH model – Upcoming

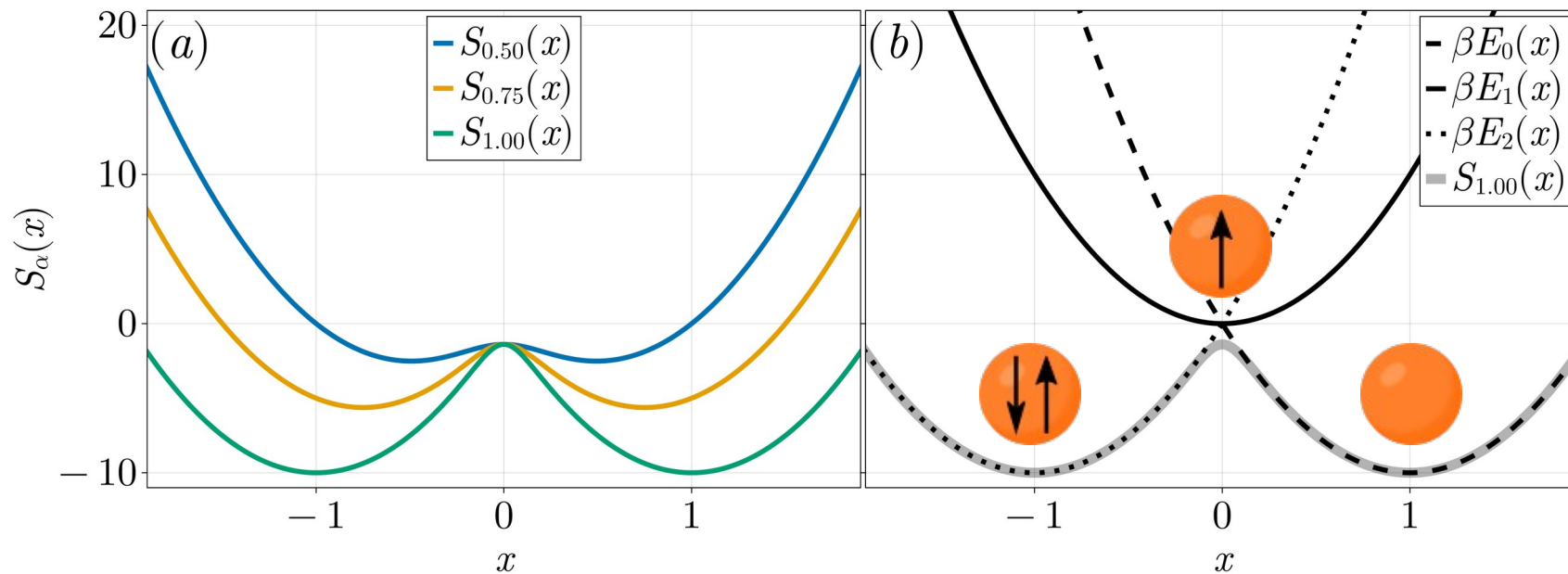
$$\begin{aligned}\hat{H}_{\text{oSSH}} = & -t \sum_{\sigma, \langle i, j \rangle} \left(c_{i, \sigma}^\dagger c_{j, \sigma} + \text{h.c.} \right) - \mu \sum_{\sigma, i} c_{i, \sigma}^\dagger c_{j, \sigma} \\ & + \sum_{\sigma, i} \left(\frac{\hat{P}_{i, \sigma}^2}{2M} + \frac{M}{2} \Omega^2 \hat{X}_{i, \sigma}^2 \right) + \alpha \sum_{\sigma, \langle i, j \rangle} \left(\hat{X}_{i, \sigma} - \hat{X}_{j, \sigma} \right) \left(c_{i, \sigma}^\dagger c_{j, \sigma} + \text{h.c.} \right)\end{aligned}$$

$$\lambda_{\text{oSSH}} = \frac{8\alpha_0^2}{M\Omega^2 W}$$

Local Trapping

- Single-site Holstein model

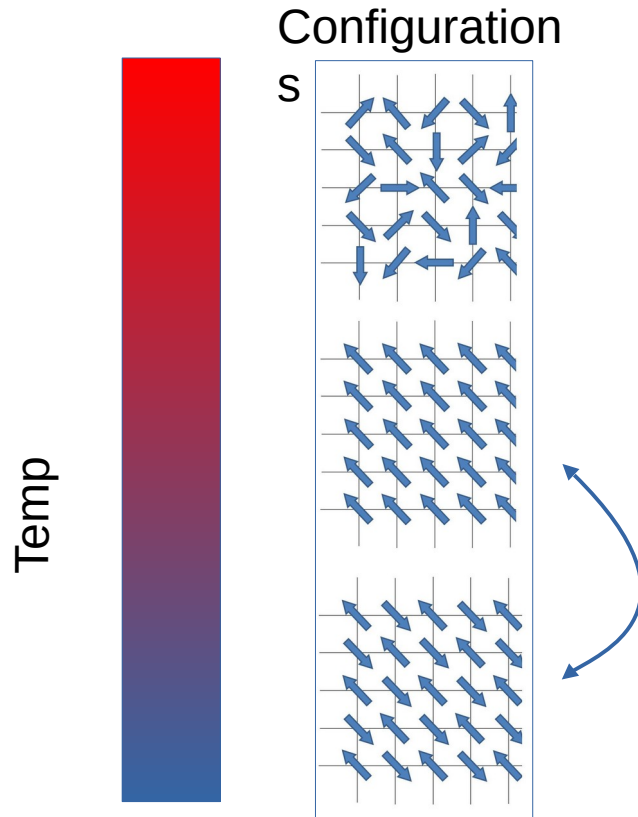
$$\hat{H} = \left(\frac{1}{2M} \hat{P}^2 + \frac{1}{2} M \Omega^2 \hat{X}^2 \right) + \alpha \sum_{\sigma} \hat{X} \left(\hat{c}_{\sigma}^{\dagger} \hat{c}_{\sigma} - \frac{1}{2} \right)$$



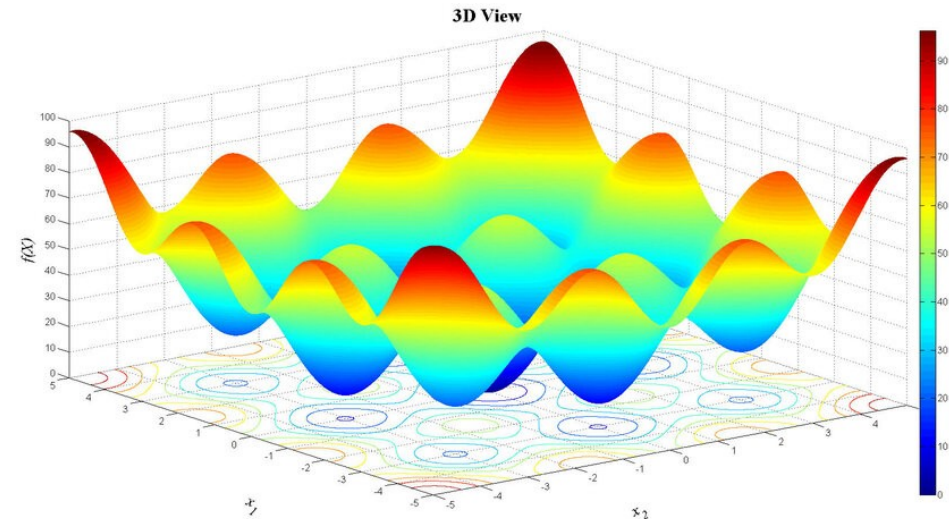
$$S_{\alpha}(x) = \beta \frac{\Omega^2}{2} x^2 - 2 \ln \cosh \left(\frac{\beta}{2} \alpha x \right) - 2 \ln 2$$

$$E_n(x) = \frac{1}{2} \Omega^2 x^2 + \alpha x (n - 1)$$

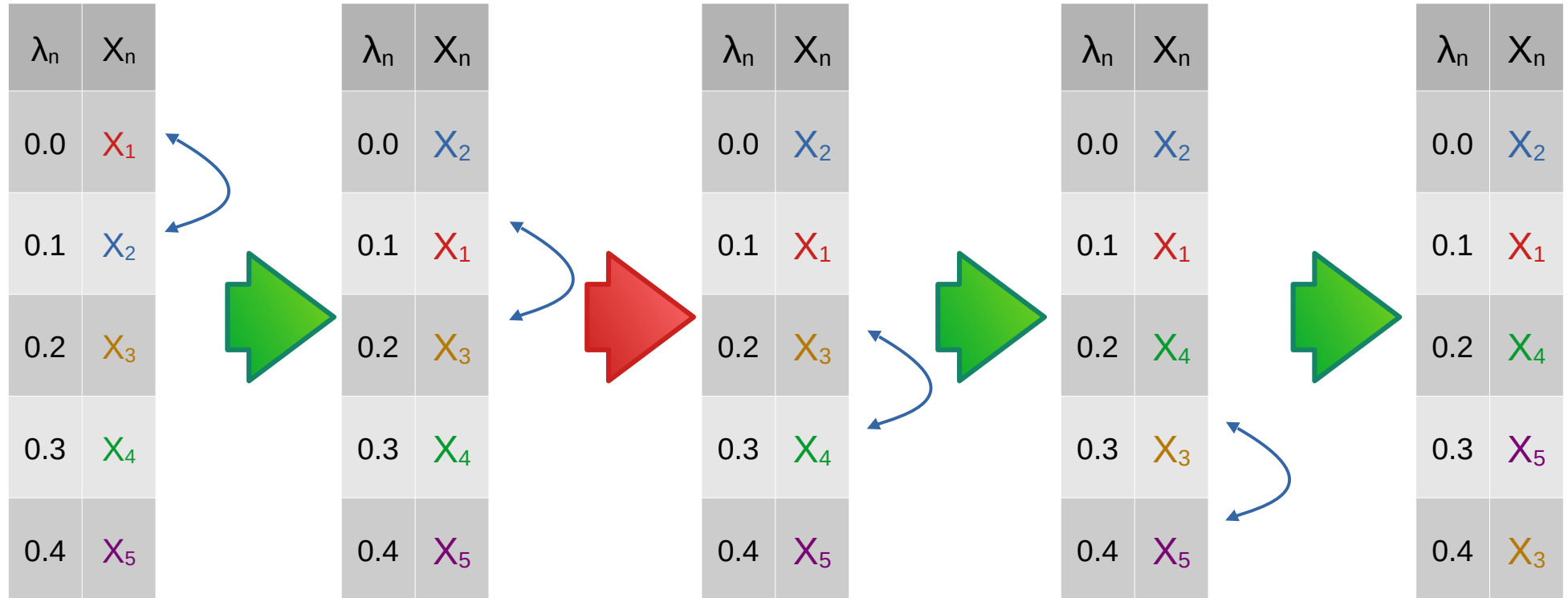
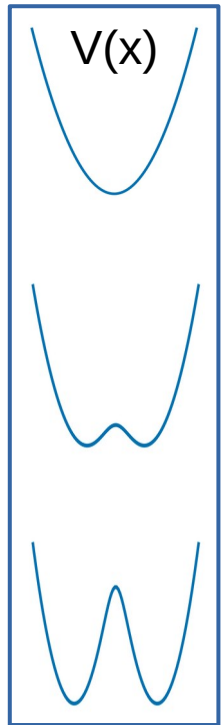
Parallel Tempering – Spin Glass



- Find ground state despite local trapping
- Temper on temperature



Quantum PT



$$P_{\text{exchange}}(X_j, X_{j'}) = \min \left[1, \frac{w_i(X_{j'})w_{i'}(X_j)}{w_i(X_j)w_{i'}(X_{j'})} \right]$$

Questions?

Parallel Tempering: Local Trapping

- Holstein Model Crossover
- Optical SSH Crossover
- Parallel Tempering

Parallel Tempering: Local Trapping

- Holstein Model Crossover
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Parallel Tempering: Local Trapping

- Holstein Model Crossover
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